# PROOFS FROM A PROGRAM: TEACHER-FREE DERIVATIONS AS MIDTRAINING DATA, AND THE DEPTH AT WHICH FORMAL NOTATION WINS

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## ABSTRACT

Synthetic pretraining data is almost always written by a language model, which bounds it by what that model knows and passes on its errors. We study a source with no model in it: a program samples a multi-hop deduction problem, solves it, and writes the solution out, so every document is correct by construction. Replacing ten percent of a 2.5 billion token midtraining mixture for Qwen2.5-7B with such derivations raises accuracy on held-out ProofWriter problems from 44.4 to 57.0 percent at inference depth 3. Replacing the same share with ordinary documents of the same length distribution leaves it at 45.4 percent. The midtrained models never reproduce the generator's notation, and their scores on general benchmarks stay within sampling error of the control. The same latent proof can be written as a formal derivation or as controlled English, and we generate both from one source so that only the notation differs. Fine-tuning OLMo-3-7B on either side of the pair, English supervision generalizes further when training proofs are at most 20 hops deep; at 25 hops the ordering reverses, with formal supervision at $75.0 \pm 9.5$ percent out-of-distribution accuracy against $17.8 \pm 4.7$ and $99.6 \pm 0.3$ percent within sixteen samples against $42.5 \pm 5.8$. Because a formal derivation is checkable, a proof checker selects a correct sample without an answer key, at $97.3 \pm 1.1$ percent single-answer accuracy against $33.8 \pm 1.4$ for English traces. In the midtraining mixture, where derivations are a small fraction of the data, English transfers at least as well as formal. The formal advantage is a property of dense, deep supervision, which is also the regime in which a model's reasoning can be verified, and we argue it is the regime worth training toward.

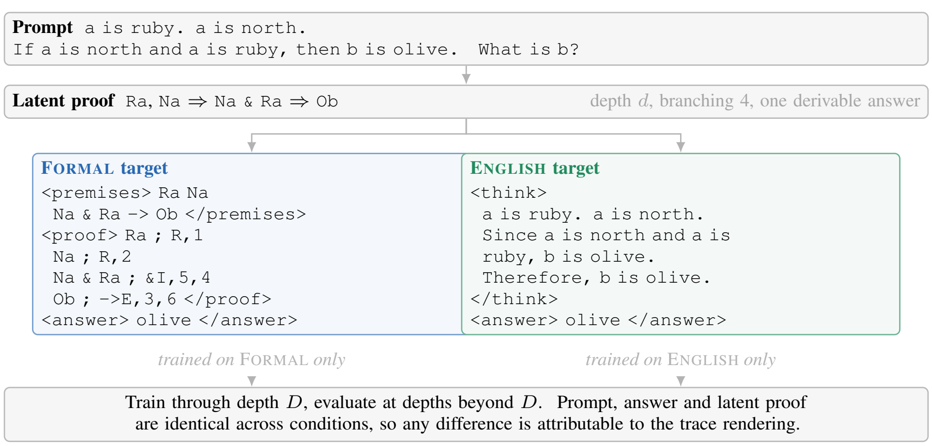


Figure 1: How the two supervised targets are produced. A generator samples a latent proof and writes it out as a formal derivation or as a controlled English explanation. The prompt, the answer and the proof graph are shared, and only the notation of the supervised target differs. Models are trained on one notation and evaluated beyond the training depth range.

# 1 INTRODUCTION

The public supply of human-written text is finite, and current projections place its exhaustion for frontier training runs within this decade (Villalobos et al., 2024). The standard remedy is synthetic text, and in nearly every published recipe a language model writes it: textbook-style corpora (Gunasekar et al., 2023; Eldan & Li, 2023), rephrased web documents (Maini et al., 2024; 2025; Su et al., 2025), and expansions of a small domain corpus by prompting (Yang et al., 2025). Such data inherits the generating model's errors and blind spots, and training on it repeatedly narrows the distribution (Shumailov et al., 2024; Dohmatob et al., 2025). Verified synthetic data avoids this in theory (Feng et al., 2025), and a few systems have trained on program-generated, mechanically checked derivations at scale (Trinh et al., 2024). Whether such data belongs in the pretraining mixture of a general language model has not been tested under controls.

A few hundred lines of Python can sample a multi-hop deduction problem, solve it, and write the solution out. The result is correct by construction, unlimited in quantity, and free of any teacher. It is also narrow and unlike the web. We test whether it is worth including. Replacing ten percent of a 2.5 billion token midtraining mixture for Qwen2.5-7B (Yang et al., 2024) with program-generated derivations raises accuracy on held-out ProofWriter problems (Tafjord et al., 2021) from 44.4 to 57.0 percent at inference depth 3 (Table 3). Replacing the same ten percent with long documents from the same corpus, matched to the derivations in length distribution, gives 45.4 percent. The gain is confined to items that require inference, it survives a correction for label bias, and none of the model's 1,000 greedy generations on the near-transfer items reproduces the generator's notation, so the model retained the deduction and not the format.

A latent proof can be rendered as a formal derivation with numbered lines and named inference rules, or as controlled English, and which rendering the program should write is the second question. We render every proof both ways from one source, so that the prompt, the answer and the proof graph are identical within a pair and only the notation differs (Figure 1). Models trained on the two sides of the pair are as close to a controlled comparison as training data permits. Prior synthetic deduction corpora render their proofs in English only (Morishita et al., 2023; 2024), and the one study that compared symbolic and English supervision on matched problems found English to generalize better (Zhou et al., 2025).

We find that both orderings occur, and that training depth decides between them. Fine-tuning OLMo-3-7B (Team OLMo et al., 2025) on 50,000 paired examples per condition, English supervision generalizes further when training proofs reach at most 20 hops. At 25 hops the ordering reverses, with formal supervision at $75.0 \pm 9.5$ percent out-of-distribution accuracy against $17.8 \pm 4.7$ (Table 1). Sixteen samples instead of one move the reversal to 15 hops, so formal supervision widens the set of problems the model can solve before it improves the first attempt, and because a formal derivation is checkable, a proof checker turns that wider set into $97.3 \pm 1.1$ percent single-answer accuracy with no answer key, against $33.8 \pm 1.4$ for English traces. The reversal is not unconditional. A spurious marker planted in training removes it, and a 32B model prefers English on the first attempt while keeping formal coverage at 98.8 percent.

In the midtraining mixture the derivations are a tenth of the data and the model is instruction-tuned afterwards, and there the English rendering transfers at least as well as the formal one, 57.0 against 50.0 percent on ProofWriter at depth 3. Read together, the three settings say that the formal advantage is a property of dense, deep supervision: a small dose of formal traces is learned less thoroughly than the same dose of prose. The dense regime is also the only one in which a trained model's reasoning can be verified by a checker, and verifiable-reward training converts sampled coverage into first-attempt accuracy (Yue et al., 2025), so we argue it is the regime worth training toward.

# 2 RELATED WORK

**Synthetic data for pretraining.** Small models pretrained on generated textbooks reach far above their size (Gunasekar et al., 2023; Eldan & Li, 2023), rephrased web text speeds pretraining severalfold (Maini et al., 2024), and the recipe has been scaled to trillions of tokens (Maini et al., 2025; Su et al., 2025) and into production mixtures (Abdin et al., 2024; Ben Allal et al., 2025; Team OLMo et al., 2025). Continued pretraining on generated expansions of a small corpus gives log-linear gains (Yang et al., 2025), and the gains from synthetic data are largest at modest mixing fractions (Kang et al., 2025). All of these use a language model as the generator. Training on model output degrades the tail of the distribution (Shumailov et al., 2024; Dohmatob et al., 2024; Alemohammad et al., 2024), at fractions as low as one percent in the scaling regime (Dohmatob et al., 2025), unless real data accumulates alongside (Gerstgrasser et al., 2024) or a verifier filters the synthetic samples (Feng et al., 2025). Our generator is a program with a built-in checker, so its output cannot carry a model's errors.

**Procedural data without a generator model.** Pretraining on structured non-linguistic data transfers to language (Papadimitriou & Jurafsky, 2020; Chiang & Lee, 2022), a large share of pretraining's benefit is reproduced by trivially simple synthetic tasks (Wu et al., 2022; Krishna et al., 2021), and pre-pretraining on formal languages lowers natural-language loss at a smaller token budget (Hu et al., 2025). Program-generated, verified geometry proofs have trained an olympiad-level prover with no human data (Trinh et al., 2024), and program-generated grade-school math with controlled dependency depth teaches a reasoning skill that generalizes to deeper problems (Ye et al., 2025). Influence functions attribute reasoning to procedural documents in pretraining (Ruis et al., 2025), midtraining helps most for domains far from web prose (Liu et al., 2025a), and upsampling reasoning-dense data late in training yields large gains at small cost (Blakeney et al., 2024; Wang et al., 2025b). Procedural generators with verifiers are standard for reinforcement learning (Stojanovski et al., 2025; Liu et al., 2025b; Chen et al., 2025a; Xie et al., 2025), and concurrent work supplies procedural corpora for pretraining (Lacombe et al., 2026; Jiang et al., 2026). None of this work places a program-generated deduction corpus into a production midtraining mixture under a length-matched control.

**Synthetic deduction corpora.** RuleTaker (Clark et al., 2020) and ProofWriter (Tafjord et al., 2021) generate rule-and-fact theories and render them in English, with depth generalization degrading beyond the training range. PrOntoQA generates problems from a first-order world model so that a chain of thought can be parsed back into a proof (Saparov & He, 2023; Saparov et al., 2023). FLD grounds a deduction corpus in a complete set of inference rules (Morishita et al., 2023), and additional training on its successor improves a 70B model on logic, mathematics and code (Morishita et al., 2024). Statistical shortcuts can satisfy such corpora in distribution and fail out of it (Zhang et al., 2023). Each of these corpora renders its proofs in English. Prolog solver traces (Feng et al., 2024) and Hilbert-style Boolean proofs (Xia et al., 2025) have been used as fine-tuning targets without a matched English arm. The closest study fine-tunes on one English and three symbolic supervision styles for the same problems and concludes that English generalizes better to long chains (Zhou et al., 2025), which matches our result at training depths up to 20 and reverses at 25.

**The language of reasoning traces.** Intermediate steps improve multi-step accuracy (Wei et al., 2022; Kojima et al., 2022; Nye et al., 2021), rationales are an efficient supervision signal (Hsieh et al., 2023; Li et al., 2023), and constant-depth transformers need serial tokens to carry out serial computation (Feng et al., 2023; Merrill & Sabharwal, 2024; Li et al., 2024b). Format governs length generalization (Anil et al., 2022; Zhou et al., 2024a;b; Hu et al., 2024). At inference time, writing reasoning as code or as a formal specification and executing it beats prose (Gao et al., 2023; Chen et al., 2023; Li et al., 2024a; Ye et al., 2023; Pan et al., 2023; Olausson et al., 2023; Lyu et al., 2023), and the formal surface helps even without execution (Puerto et al., 2024). As training data, program-format solutions beat chain-of-thought for small math models because programs can be executed and filtered (Zhu et al., 2024; Yue et al., 2024). Prose rationales can misrepresent what drove an answer (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025b), which a checked derivation cannot. None of these studies holds the latent proof fixed while varying the notation.

**Coverage, verification and verifiable rewards.** Coverage grows with the number of samples and converts into accuracy only where an automatic verifier exists (Brown et al., 2024), with verifiers trained (Cobbe et al., 2021; Lightman et al., 2024) or replaced by agreement (Wang et al., 2023). Reinforcement learning from verifiable rewards mostly sharpens a model toward what it could already sample (Yue et al., 2025; Shao et al., 2025; Agarwal et al., 2025), supervised fine-tuning expands the set of correct trajectories where reinforcement learning compresses the incorrect ones (Matsutani et al., 2026), and prolonged training can widen coverage again (Liu et al., 2025c; Cheng et al., 2025). In formal theorem proving the checker is the reward (Xin et al., 2025; Lin et al., 2025; Wang et al., 2025a; Hubert et al., 2025).

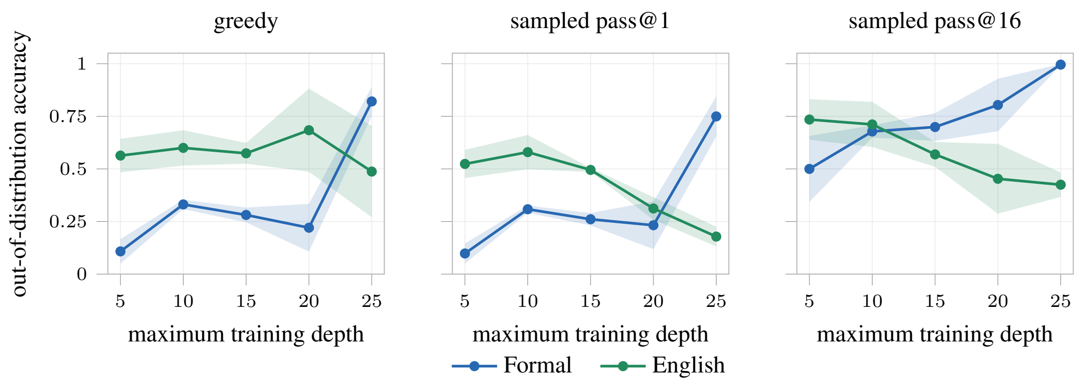


Figure 2: Out-of-distribution answer accuracy on BRANCHPROOF against maximum training depth, for three decoding rules. Shaded regions give one standard deviation over three seeds. Greedy decoding and pass@1 place the crossover at maximum training depth 25. Sampling with sixteen draws places it at 15, ten depth levels earlier.

## 3 Paired Derivations from One Generator

Each example begins with a latent directed acyclic proof. The generator emits a prompt containing ground facts, rules, a query and distractor branches, then traverses the supporting path to produce the answer. A rendering function writes that path in one of two notations. The FORMAL rendering declares constants and predicates and applies named inference rules to numbered lines. The ENGLISH rendering expresses the same substitutions and conclusions, line for line, in controlled sentences. Both targets end with the same answer field, and the prompt is English in both conditions.

BRANCHPROOF composes Horn-style inferences in which every step offers four locally plausible branches, exactly one derivable from the current state, with a fresh constant introduced at every layer. The generator computes the full Horn closure of each example and accepts it only when exactly one candidate answer is derivable, which an independent solver confirms over 3,000 instances (Appendix A). The branching factor is 4 and the distractor ratio 0.5 in every experiment, depths are sampled round-robin, and corpus seeds are disjoint across training, evaluation and sampling. Appendices B to D give the grammar, the generator, the validators and complete paired targets.

The controlled study fine-tunes OLMo-3-7B (Team OLMo et al., 2025) with LoRA on 50,000 examples per condition, with maximum training depth $D \in \{5, 10, 15, 20, 25\}$ and three seeds per cell. Evaluation covers depths 1 to 50 with 32 prompts per depth, sixteen samples at temperature 1.0 per prompt and one greedy decode, and the unbiased pass@k estimator (Chen et al., 2021). Depths above $D$ are out of distribution. Both notations receive the same 7,168-token generation limit inside a 16,384-token window, and a tokenizer audit confirms that every gold target fits in both renderings. Joint metrics require a correct answer and a mechanically valid proof, with formal traces checked directly and English traces checked after a deterministic translation into the same engine. Appendices E and F give the configuration and the metric definitions.

## 4 Training Depth Decides Which Notation Is Better

Figure 2 and Table 1 report out-of-distribution answer accuracy as a function of the maximum training depth $D$. Through $D = 20$ the ENGLISH condition leads under every decoding rule. At $D = 25$ the FORMAL condition reaches $75.0 \pm 9.5$ percent pass@1 against $17.8 \pm 4.7$, and $99.6 \pm 0.3$ percent pass@16 against $42.5 \pm 5.8$. Greedy decoding gives the same ordering as pass@1 at every depth.

pass@1 crosses at $D = 25$ and pass@16 at $D = 15$, where the FORMAL condition reaches 69.9 percent against 56.9. Between the two crossings the formally supervised model can construct a correct derivation within sixteen attempts but not on the first, so the set of solvable problems widens ten depth levels before single-sample reliability improves. A study reporting greedy accuracy alone would describe $D = 15$ and $D = 20$ as showing no formal advantage. The two conditions also

Table 1: Out-of-distribution answer accuracy on BRANCHPROOF in percent, mean and standard deviation over three seeds, by maximum training depth $D$. The better condition per cell is in bold. Greedy decoding and pass@1 cross at $D = 25$; pass@16 crosses at $D = 15$.

| decoding | condition | $D = 5$ | $D = 10$ | $D = 15$ | $D = 20$ | $D = 25$ |
|---|---|---|---|---|---|---|
| greedy | FORMAL | $10.8 \pm 5.9$ | $33.1 \pm 2.2$ | $28.1 \pm 3.5$ | $22.0 \pm 11.3$ | $\mathbf{82.1 \pm 7.0}$ |
| greedy | ENGLISH | $\mathbf{56.3 \pm 8.0}$ | $\mathbf{60.0 \pm 8.4}$ | $\mathbf{57.4 \pm 5.0}$ | $\mathbf{68.4 \pm 19.8}$ | $48.8 \pm 21.7$ |
| pass@1 | FORMAL | $9.8 \pm 4.8$ | $30.8 \pm 1.7$ | $26.1 \pm 2.9$ | $23.2 \pm 11.3$ | $\mathbf{75.0 \pm 9.5}$ |
| pass@1 | ENGLISH | $\mathbf{52.4 \pm 6.7}$ | $\mathbf{58.0 \pm 8.1}$ | $\mathbf{49.5 \pm 1.0}$ | $\mathbf{31.2 \pm 5.4}$ | $17.8 \pm 4.7$ |
| pass@16 | FORMAL | $50.0 \pm 15.7$ | $67.8 \pm 2.9$ | $\mathbf{69.9 \pm 6.6}$ | $\mathbf{80.4 \pm 12.5}$ | $\mathbf{99.6 \pm 0.3}$ |
| pass@16 | ENGLISH | $\mathbf{73.5 \pm 9.6}$ | $\mathbf{71.2 \pm 10.8}$ | $56.9 \pm 5.9$ | $45.3 \pm 16.6$ | $42.5 \pm 5.8$ |

respond to deeper training proofs in opposite directions: ENGLISH pass@16 falls from 73.5 percent at $D = 5$ to 42.5 at $D = 25$, and FORMAL pass@16 rises from 50.0 to 99.6. Neither notation is better in general. The ordering flips with training depth, and where it flips depends on the decoding rule.

**Supervision quantity.** Formal targets are longer than English targets, so the comparison could measure the number of supervised tokens. Equating supervised target tokens across conditions by subsampling gives $41.0 \pm 20.1$ percent pass@1 and $69.8 \pm 34.9$ percent pass@16 for the token-matched ENGLISH condition at $D = 25$ (Table 2), against $99.6 \pm 0.3$ pass@16 for FORMAL. The ordering is unchanged. The variance of this control is large, and we report it as bounding the explanation and not as a precise estimate.

### 4.1 A CHECKER SELECTS THE CORRECT SAMPLE WITHOUT LABELS

A model whose sixteen samples contain a correct proof is only useful if that proof can be found without the answer. For formal derivations it can, since validity is decided by the same engine that generated the problem. Drawing sixteen samples, checking each in order and returning the answer of the first one the checker accepts gives $97.3 \pm 1.1$ percent single-answer accuracy at $D = 25$, against a joint pass@16 ceiling of $98.1 \pm 0.5$ percent. The $0.8$ point gap is the cost of derivations the checker accepts that reach a wrong answer. No reward model and no gold labels enter the procedure. The same selector applied to translated ENGLISH traces recovers $33.8 \pm 1.4$ percent, mainly because long English derivations fail to terminate inside the generation budget, so there is no checkable derivation to accept.

### 4.2 WHERE THE REVERSAL HOLDS AND WHERE IT DOES NOT

Table 2 collects controls run on the same generator at $D = 25$.

**Shortcuts.** A spurious marker that identifies the active branch on 80 percent of training examples and is removed at test time (Geirhos et al., 2020) reverses the comparison. The ENGLISH condition is barely affected, at $99.4 \pm 0.9$ percent pass@16 and 76.1 percent at depth 50, while the FORMAL condition falls to $87.3 \pm 7.5$ and to 11.7 percent at depth 50. The formal advantage is therefore not a general robustness to distribution shift. A model trained on rigid syntax binds more readily to a rigid spurious feature.

**Both notations in one model.** Concatenating the two renderings into one target fails, at $0.5 \pm 0.4$ and $0.2 \pm 0.1$ percent pass@1, because the model copies one or both long traces and exhausts its budget. Separate examples with an output-mode tag reach $50.3 \pm 6.4$ percent pass@1 in formal mode, below single-notation training.

**Base model.** On Qwen2.5-7B the conditions are within variance of each other, at $69.9 \pm 11.1$ against $74.9 \pm 7.0$ percent pass@1. On OLMo-3-32B ENGLISH leads at $98.9 \pm 0.8$ against $63.9 \pm 5.1$ percent pass@1, while FORMAL pass@16 stays at $98.8 \pm 0.9$. At 32B the formal deficit is one of

Table 2: Out-of-distribution answer accuracy in percent at maximum training depth 25, mean and standard deviation over three seeds. All rows use the same generator and evaluation as Table 1.

| condition | pass@1 | pass@16 |
|---|---|---|
| *reference* | | |
| FORMAL | $75.0 \pm 9.5$ | $99.6 \pm 0.3$ |
| ENGLISH | $17.8 \pm 4.7$ | $42.5 \pm 5.8$ |
| *supervised tokens equated* | | |
| ENGLISH, token-matched | $41.0 \pm 20.1$ | $69.8 \pm 34.9$ |
| *spurious marker in training, absent at test* | | |
| FORMAL | $43.1 \pm 7.7$ | $87.3 \pm 7.5$ |
| ENGLISH | $78.5 \pm 14.9$ | $99.4 \pm 0.9$ |
| *both notations in one model* | | |
| concatenated, FORMAL first | $0.5 \pm 0.4$ | $7.1 \pm 5.5$ |
| concatenated, ENGLISH first | $0.2 \pm 0.1$ | $2.3 \pm 1.6$ |
| mode-conditioned, FORMAL mode | $50.3 \pm 6.4$ | $87.1 \pm 3.9$ |
| mode-conditioned, ENGLISH mode | $18.1 \pm 2.0$ | $24.8 \pm 3.7$ |
| *other base models* | | |
| Qwen2.5-7B, FORMAL | $69.9 \pm 11.1$ | $77.3 \pm 5.2$ |
| Qwen2.5-7B, ENGLISH | $74.9 \pm 7.0$ | $76.0 \pm 5.6$ |
| OLMo-3-32B, FORMAL | $63.9 \pm 5.1$ | $98.8 \pm 0.9$ |
| OLMo-3-32B, ENGLISH | $98.9 \pm 0.8$ | $99.8 \pm 0.3$ |
| OLMo-3-32B conditioned, FORMAL mode | $85.0 \pm 8.7$ | $99.8 \pm 0.3$ |
| OLMo-3-32B conditioned, ENGLISH mode | $77.2 \pm 11.7$ | $88.8 \pm 7.7$ |

probability mass on the first sample, not of coverage. Mode conditioning at 32B lifts the formal mode to $85.0 \pm 8.7$ percent pass@1. Across the three base models, FORMAL pass@16 is at or above ENGLISH pass@16 in every $D = 25$ cell except Qwen2.5-7B, where the two are within one standard deviation.

## 5 MIDTRAINING A 7B MODEL ON DERIVATIONS FROM A PROGRAM

The controlled study supplies dense supervision on one task, and the practical question is whether a small share of such data improves an ordinary model when placed in an ordinary mixture, on benchmarks the generator never saw. We test this by midtraining Qwen2.5-7B (Yang et al., 2024) on the Dolmino mixture (Team OLMo et al., 2025) with a fixed fraction replaced.

### 5.1 DESIGN

Five conditions share every hyperparameter, every data path outside the replaced slice, and a token budget of 2,500,853,760 training tokens, reached as 2,385 steps of global batch 128 at sequence length 8,192. They differ in the ten percent of loss tokens that is replaced.

1. CONTROL trains on the unmodified mixture.
2. LONGDOC replaces ten percent with long documents drawn from the same mixture, selected to match the token-length histogram of the FORMAL corpus.
3. FORMAL replaces ten percent with formal derivations generated at depths 1 to 25.
4. ENGLISH replaces ten percent with the English rendering of the same latent proofs.
5. CONDENSED replaces ten percent with a formal rendering of the same latent proofs that states the theory once, 2.6 times shorter.

The FORMAL document states its theory twice, as the English prompt and again as formal premises, and then gives the proof; the ENGLISH document states the theory once, in English. Formal premises

Table 3: Greedy exact match on ProofWriter by inference depth, in percent, 500 items per depth, one training run per condition at instruction-tuning seed 3407. Depth 0 requires lookup only. The right-hand block pools depths 1 to 5 (2,000 items) and splits accuracy by gold label; *two-class* is accuracy on the 1,527 items whose gold label is true or false.

| | inference depth | | | | | pooled depths 1–5 | | | |
|---|---|---|---|---|---|---|---|---|---|
| condition | 0 | 1 | 2 | 3 | 5 | pred. true | gold true | gold false | two-class |
| CONTROL | 44.2 | 32.8 | 44.0 | 44.4 | 46.0 | 66.3 | 70.4 | 38.5 | 54.4 |
| LONGDOC | 45.8 | 32.2 | 44.4 | 45.4 | 44.0 | 65.9 | 69.5 | 38.6 | 54.0 |
| FORMAL | 48.0 | 35.8 | 49.4 | 50.0 | 50.2 | 56.4 | 66.5 | 54.2 | 60.3 |
| ENGLISH | 48.8 | 38.8 | 56.2 | 57.0 | 56.4 | 49.5 | 68.3 | 68.0 | 68.2 |
| CONDENSED | 46.0 | 34.6 | 46.0 | 48.0 | 47.4 | 56.8 | 64.0 | 50.9 | 57.4 |

are about 1.5 times more compact than controlled English for the same theory, so the two documents are length-matched by that offset, and CONDENSED is the formal document with the theory stated once. LONGDOC separates two explanations of any gain, since derivations are long documents and long documents change what a model learns about carrying state across a window. Its median length is 3,712 tokens against 3,749 for FORMAL, and all thirty populated 250-token bins agree within three percent. The three derivation corpora contain 72,000 documents each and render the same latent proofs. A document-preserving packer never splits a document across windows, and a decoded-batch audit refuses to start training if any window contains a split document or an unmasked padding token; zero documents were split or excluded in any condition. Because the replaced slice is fixed in tokens, CONDENSED traverses its proof set 2.31 times where FORMAL and ENGLISH traverse theirs 0.90 and 0.89 times, so its comparison with ENGLISH is one of a compact rendering seen more than twice against a verbose one seen once (Appendix I).

All five checkpoints receive the same instruction tuning on 100,000 examples of a public instruction corpus at seed 3407, and a second replicate at seed 3408 is reported where available. Evaluation uses two families. The external family is ProofWriter under the open-world assumption at inference depths 0, 1, 2, 3 and 5, with 500 items per depth, which the generator never produced. The near-transfer family asks for a written derivation on held-out problems from the generator at depths 5 to 25, with 200 items per depth. A ten-task general suite and three multi-hop question-answering tasks check for regressions. Sampled measurements draw sixteen completions at temperature 0.8 on the prompts logged by the greedy run. Appendix I gives every setting.

### 5.2 RESULTS

**External transfer.** At depth 0 of ProofWriter, which requires lookup and no inference, the five conditions fall between 44.2 and 48.8 percent greedy exact match (Table 3). From depth 2 onward ENGLISH leads CONTROL by 10 to 13 points and FORMAL leads it by 4 to 6 points, and LONGDOC stays within 2 points of CONTROL at every depth. The gain therefore follows from the deduction in the replaced slice and not from its length, and it appears only where inference is required.

The right-hand block of Table 3 splits the pooled items by gold label. CONTROL and LONGDOC predict *true* on 66 percent of items. The derivation conditions predict it on 50 to 57 percent, their accuracy on gold-true items stays within 6 points of CONTROL, and their accuracy on gold-false items rises from 38.5 to between 50.9 and 68.0 percent, so the gain is a gain in rejecting false claims and not a shift of the prior. No condition answers the *unknown* class correctly on more than 1.3 percent of its 473 items, so the result is a two-class deduction gain.

**Near transfer.** On held-out problems from the generator, prompted to write a derivation before answering, CONTROL scores between 3.5 and 11.0 percent greedy exact match across depths and LONGDOC between 7.5 and 15.0. Every derivation condition is far above both: ENGLISH between 24.5 and 52.5 percent, FORMAL between 17.5 and 31.5, and CONDENSED between 18.5 and 34.5 (Table 4). The replaced slice teaches a capability the control mixture does not supply. Under greedy decoding ENGLISH is ahead of FORMAL at every depth, by 3.0 to 21.0 points.

Table 4: Near-transfer accuracy in percent by inference depth and decoding rule, 200 items per depth, sixteen samples at temperature 0.8. Greedy values come from the instruction-tuned checkpoints at seed 3407.

| decoding | condition | d5 | d10 | d15 | d20 | d25 |
|---|---|---|---|---|---|---|
| greedy | CONTROL | 5.0 | 3.5 | 11.0 | 9.0 | 6.5 |
| greedy | LONGDOC | 7.5 | 11.0 | 12.0 | 15.0 | 12.0 |
| greedy | FORMAL | 31.5 | 17.5 | 27.5 | 21.5 | 19.5 |
| greedy | ENGLISH | 52.5 | 34.5 | 30.5 | 24.5 | 29.5 |
| greedy | CONDENSED | 34.5 | 28.5 | 18.5 | 20.5 | 24.0 |
| pass@16 | CONTROL | 58.0 | 55.5 | 58.5 | 60.0 | 53.0 |
| pass@16 | LONGDOC | 63.0 | 63.5 | 61.0 | 63.5 | 56.0 |
| pass@16 | FORMAL | 97.0 | 85.5 | 79.0 | 74.0 | 71.5 |
| pass@16 | ENGLISH | 95.5 | 91.5 | 84.5 | 78.0 | 82.5 |
| pass@16 | CONDENSED | 89.5 | 89.0 | 80.0 | 81.5 | 77.5 |

The proportion of greedy generations that reach an answer field is 64.5 to 78.5 percent for CONTROL, 79.5 to 92.5 for LONGDOC, 94.0 to 99.5 for FORMAL, and 100 for ENGLISH, so models that never saw a derivation in midtraining often stop before concluding.

**Sampled decoding.** Sampling raises every condition and preserves the greedy ordering (Table 4). Pooled over depths, pass@16 is 57.0 percent for CONTROL, 61.4 for LONGDOC, 81.4 for FORMAL, 86.4 for ENGLISH and 83.5 for CONDENSED, and mean single-sample accuracy is 6.9, 7.7, 20.6, 27.4 and 17.5. A paired bootstrap over the 1,000 documents (Appendix J) gives ENGLISH minus FORMAL at $+5.0$ points of pass@16, interval $[2.1, 7.9]$, and $+6.8$ points of single-sample accuracy, interval $[5.2, 8.3]$; FORMAL minus LONGDOC is $+20.0$ points of pass@16, interval $[16.3, 23.5]$. CONDENSED sits between the two notations in coverage, 2.9 points below ENGLISH, interval $[-0.1, 5.8]$, and 9.9 points below it in single-sample accuracy. The sampled crossover of the controlled study, where formal coverage overtook English at $D = 15$, does not appear here. At a tenth of the mixture the formal rendering is behind the English rendering on the first sample and on coverage alike.

**General benchmarks.** On nine general benchmarks and three long-context multi-hop tasks under standard prompting, the five conditions are within 1.3 points of each other on GSM8K, MATH-500, BBH, MMLU, HellaSwag, WinoGrande and PIQA, with GSM8K between 78.6 and 79.8 percent and MMLU between 69.0 and 69.8 (Table 6). The largest spread is on LogiQA, where CONTROL scores 31.2 and the derivation conditions 26.6 to 28.1 on 651 items, and on 2WikiMultihopQA, 34.3 to 38.9 on 200 items, in both cases with CONTROL at or near the top and within two standard errors of the others. Replacing ten percent of the mixture does not help tasks that do not ask for a derivation, and any cost on them is at the level of sampling error.

**The midtrained models do not write formal notation.** None of the 1,000 greedy near-transfer generations of the FORMAL arm contains any marker of the training rendering, and the same holds for every other condition. After instruction tuning all five models answer in prose. The gain reported above is therefore not an effect of adopting formal notation, and the checker-based selection of Section 4.1 does not apply at this scale, since no formal derivation is emitted for a checker to accept.

### 5.3 STATISTICAL STATUS

Every midtrained checkpoint was instruction-tuned twice under different seeds. Over the fifty condition and task pairs of the deduction suite the mean absolute difference between seeds is $0.015$ and the largest is $0.050$, implying a standard deviation of $0.021$ for a difference between two arms, and every contrast above survives: at inference depth 3, ENGLISH minus CONTROL is $+12.6$ and $+13.6$ points across the two seeds, FORMAL minus CONTROL is $+5.6$ and $+6.6$, and LONGDOC minus CONTROL is $+1.0$ and $+1.4$ (Appendix J). The replicate does not address variation between midtraining runs, which were single, so a different midtraining seed could reorder arms separated by a few points. That applies to FORMAL against CONDENSED and not to either against CONTROL or LONGDOC.

## 6 DISCUSSION

**What transfers is the deduction.** Both renderings raise ProofWriter accuracy, the length-matched control does not, and the trained models write neither notation, so what survives instruction tuning is the content of a program-generated derivation and not its surface form. That is the property one wants from a source which costs nothing to run and cannot carry a generating model's errors.

**The formal advantage is a property of dense, deep supervision.** Across the settings studied, formal supervision wins the first attempt in exactly one: 50,000 examples on a single task, training proofs through 25 hops, on a 7B model. It wins coverage more broadly, at $D \geq 15$ and at 32B. It loses the first attempt when training proofs are shallow, when a shortcut is available, when the model is larger, and when the derivations are a tenth of a midtraining mixture, where it is behind on coverage as well. The formal rendering is further from the pretraining distribution than controlled English, so a given dose of it is learned less thoroughly and the model falls back on prose-shaped behaviour on the first sample. The same dose of English is learned more thoroughly but produces a narrower distribution over derivations at depth, which is what the falling ENGLISH pass@16 curve in Table 1 measures.

**Why the dense regime is the one to train toward.** In the dense regime almost every held-out problem has a correct formal derivation within sixteen samples, at depths beyond the training range and with distractor branches at every step, and that derivation can be found by a checker with no labels. A system that samples and verifies wants exactly these two properties, and prose supervision supplies neither. Verifiable-reward training principally reweights a model toward completions it can already sample (Yue et al., 2025; Matsutani et al., 2026), so the coverage a model has after supervised training bounds what that stage can reach, and in the dense regime formal supervision moves that bound from $42.5$ to $99.6$ percent. Ten percent of a mixture is not enough to make a model write derivations a checker can accept. The next step is a larger share, or an instruction stage that preserves the notation, followed by verifiable-reward training on both notations. We have not run that stage, and our results establish only that the ingredients it needs are present after dense formal supervision and absent after prose supervision.

## 7 LIMITATIONS

The controlled study uses LoRA fine-tuning on one model family at 7B and 32B and a second at 7B, and the midtraining study uses a third, whose depth thresholds are inherited from OLMo-3-7B rather than measured on it. The generator produces one family of Horn-style deduction problems without arithmetic, ambiguity or retrieval, and the depth axis is proof depth alone. Proof validity for English traces depends on a deterministic translation into the proof engine that the formal condition does not need, so answer accuracy is reported alongside every joint metric. CONDENSED sees each proof 2.6 times more often than the other derivation conditions, and ProofWriter's *unknown* class is unsolved by every model. An earlier configuration of the midtraining study, at sequence length 4,096 with a five percent share and derivations capped at depth 14, produced no measurable transfer; it differs in window, share, rendering and evaluation suite and is not a controlled comparison (Appendix K).

## 8 CONCLUSION

A program that writes out its own deductions is a usable source of midtraining data. At ten percent of a 2.5 billion token mixture it raises held-out deduction accuracy by up to 13 points, against a length-matched control that raises nothing and with general benchmarks untouched. How the program writes matters, and the answer depends on how much of its output the model sees. Trained densely through 25 hops, formal notation solves nearly every held-out problem within sixteen samples and lets a checker pick a correct one without labels; at a tenth of a mixture, English transfers at least as well. Formal notation wins in the regime where reasoning can be verified, which is a reason to train toward that regime.

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## A Answer-Uniqueness Validation of the Generator

Every metric in this paper compares a generated answer against one labelled answer, which is only meaningful if the labelled answer is the sole one derivable from the premises. Each example introduces fresh constants `c0` through `c{depth}`, so no constant is reused across layers and no two branches can derive the same atom through reuse of a state predicate. A preflight check computes the full Horn closure of 1,000 training instances and 2,000 evaluation instances with a solver independent of the generator and records the number of candidate final states derivable in each. The unique-answer rate is 1.0, the maximum derived-candidate count is 1, there are no generation failures, and answer positions are balanced across the four candidates.

Token headroom is verified over the same instances. Evaluation uses a 16,384-token context with a 7,168-token generation cap applied identically to both notations. Across every validation row and every depth the minimum measured headroom is 494 tokens for generation and 2,788 tokens for the full context, so neither notation is truncated at any depth reported in this paper.

## B Trace Grammar and Validation

Formal traces contain four blocks, constants, predicates, premises and proof lines, followed by a conclusion and an answer. A proof line contains a formula and a rule justification such as premise retrieval or implication elimination. The checker parses the generated blocks and verifies that each proof line is licensed by earlier lines and premises. English traces use controlled sentences for the same premises and proof steps, and a deterministic translator maps those sentences back into the checker. The operational BranchProof formal surface is:

```
<formal>
<constants> CONST_DEF+ </constants>
<predicates> PRED_DEF+ </predicates>
<premises> FORMULA+ </premises>
<proof> PROOF_LINE+ </proof>
<conclusion> ATOM </conclusion>
<answer> LABEL </answer>

CONST_DEF  ::= CONST "=" LABEL
PRED_DEF   ::= PRED "x:" "x is" LABEL
ATOM       ::= PRED CONST
FORMULA    ::= ATOM | ATOM "&" ATOM "->" ATOM | ATOM "->" ATOM
PROOF_LINE ::= FORMULA "; R," REF
             | ATOM "; ->E," REF "," REF
             | ATOM "&" ATOM "; &I," REF "," REF
```

References must point to earlier proof lines or listed premises, so a line cannot justify a future step.

## C Generator and Evaluation Mechanics

The generator samples a target path, assigns labels to constants and predicates, then adds distractor branches that remain locally coherent but do not derive the queried conclusion. In pseudocode:

```
sample target answer label
sample initial object state
for depth step in 1..D:
    sample active branch rule(s)
    derive next object state
    add coherent distractor rules for inactive branches
render one prompt
render the same latent proof as formal and English targets
```

Answer extraction searches for the final `<answer>` field. If no well-formed final answer is present the generation is counted incorrect, even when earlier text contains the gold label. Formal validation parses block boundaries, declarations, formulas, proof-line references and rule applications. Common failures are malformed block structure, undeclared symbols, references to unavailable lines, applying implication elimination to the wrong antecedent, and stopping before the queried conclusion. English validation adds one failure mode: the controlled sentence must translate back into a formal step before the checker can evaluate it.

## D Paired Target Examples

The prompt and the answer are identical within each pair. Full experimental examples contain more distractor premises and longer derivations.

**Example 1.**

```
Prompt:
a is ruby. a is north.
If a is north and a is ruby, then b is olive.
What is b?
```

**Formal target**

```
<formal>
<constants> a = a b = b </constants>
<predicates> Rx: x is ruby Nx: x is north
Ox: x is olive </predicates>
<premises> Ra Na Na & Ra -> Ob </premises>
<proof> Ra ; R,1
Na ; R,2
Na & Ra ; &I,5,4
Ob ; ->E,3,6 </proof>
<conclusion> Ob </conclusion>
<answer> olive </answer>
```

**English target**

```
<think>
a is ruby. a is north.
Since a is north and a is ruby, b is olive.
Therefore, b is olive.
</think>
<answer> olive </answer>
```

**Example 2.**

```
Prompt:
a is cobalt. a is east.
If a is east and a is cobalt, then b is maple.
If b is maple, then b is south.
If b is south and b is maple, then c is cedar.
What is c?
```

**FORMAL target**

```
<formal>
<constants> a = a b = b c = c </constants>
<predicates> Cx: x is cobalt Ex: x is east
Mx: x is maple Sx: x is south Dx: x is cedar </predicates>
<premises> Ca Ea
Ea & Ca -> Mb
Mb -> Sb
Sb & Mb -> Dc </premises>
<proof> Ca ; R,1
Ea ; R,2
Ea & Ca ; &I,7,6
Mb ; ->E,3,8
Sb ; ->E,4,9
Sb & Mb ; &I,10,9
Dc ; ->E,5,11 </proof>
<conclusion> Dc </conclusion>
<answer> cedar </answer>
```

**ENGLISH target**

```
<think>
a is cobalt. a is east.
Since a is east and a is cobalt, b is maple.
Since b is maple, b is south.
Since b is south and b is maple, c is cedar.
Therefore, c is cedar.
</think>
<answer> cedar </answer>
```

# E Controlled-Study Training and Evaluation Details

All runs use materialized datasets so that the two conditions see the same prompts and answers. Each train-depth subset contains 50,000 examples and the online validation subset 256. Runs train for 10,000 optimizer steps with LoRA adapters on the query, key, value, output, up, down and gate projections, rank 16, alpha 32, dropout 0.05, learning rate $10^{-4}$, per-device batch size 1 and gradient accumulation 1 unless a control states otherwise, in bfloat16 with gradient checkpointing.

Evaluation uses vLLM with top-$p$ 0.95, temperature 1.0, sixteen samples per prompt and one greedy decode, at depths $\{1, 2, 5, 10, 12, 15, 18, 20, 25, 30, 35, 40, 45, 50\}$ with 32 prompts per depth per seed. The out-of-distribution band for a run with maximum training depth $D$ is every evaluated depth above $D$; for $D = 25$ it contains $5 \times 32 = 160$ prompts per seed. Reported uncertainty is the standard deviation over three seeds.

# F Metric Definitions

Answer accuracy is computed after extracting the final answer field. pass@k uses the unbiased estimator of Chen et al. (2021) over the sixteen samples. Formal joint validity requires a correct answer and a mechanically valid formal trace. English joint validity requires a correct answer and a successful translation of the trace into a valid formal proof. Checker selection draws the sixteen samples in a fixed order, accepts the first whose derivation the checker validates, and scores its answer; if none is accepted the item is scored incorrect. The selector never sees the gold answer.

# G Length and Token-Budget Controls

Generation budgets are identical by construction, at a 7,168-token limit inside a 16,384-token window, and the tokenizer audit of Appendix A confirms that every gold target fits in both notations. Supervised token counts are equated in the token-matched run of Table 2 by subsampling the longer condition until the number of target tokens per condition is equal. The LONGDOC condition of Section 5 supplies a third control at midtraining scale.

# H Boundary Condition Details

**Shortcut.** A spurious schema marker identifies the active branch on 80 percent of training examples and is removed at test time. Out-of-distribution pass@1 is $43.1 \pm 7.7$ percent for FORMAL and $78.5 \pm 14.9$ for ENGLISH; at depth 50 the two are $11.7$ and $76.1$ percent.

**Mode conditioning.** Concatenating both renderings into one target gives $0.5 \pm 0.4$ percent out-of-distribution pass@1 with the formal rendering first and $0.2 \pm 0.1$ with the English rendering first; generations copy one or both traces and exhaust the budget. Mode-conditioned training places an output-mode tag on separate examples. At 7B it reaches $50.3 \pm 6.4$ percent pass@1 in formal mode and $18.1 \pm 2.0$ in English mode; at 32B, $85.0 \pm 8.7$ and $77.2 \pm 11.7$. Raw generations separate cleanly by mode.

**Capacity.** Single-notation training gives out-of-distribution pass@1 of $69.9 \pm 11.1$ formal against $74.9 \pm 7.0$ English on Qwen2.5-7B, and $63.9 \pm 5.1$ against $98.9 \pm 0.8$ on OLMo-3-32B. Formal pass@16 at 32B is $98.8 \pm 0.9$.

# I MIDTRAINING CONFIGURATION

Table 5: Rendered document lengths under the Qwen2.5-7B tokenizer, 72,000 documents per corpus, and packing statistics at sequence length 8,192.

| corpus | p50 | mean | p99 | max | windows | real tokens | packing eff. |
|---|---|---|---|---|---|---|---|
| FORMAL | 3,749 | 3,856 | 7,673 | 7,727 | 35,492 | 277,732,923 | 0.9551 |
| ENGLISH | 3,819 | 3,907 | 7,744 | 7,851 | 35,979 | 281,372,725 | 0.9545 |
| CONDENSED | 1,450 | 1,500 | 3,007 | 3,014 | 13,311 | 108,099,330 | 0.9912 |
| LONGDOC | 3,712 | 3,832 | n/a | 7,750 | n/a | n/a | n/a |

**Optimisation.** Every condition trains Qwen2.5-7B for 2,385 steps at sequence length 8,192 with a global batch of 128 sequences, formed as micro-batch 2 with gradient accumulation 32 over eight A100-80GB devices, for 2,500,853,760 tokens per condition. The learning rate is $10^{-5}$ with minimum $10^{-6}$, 128 warmup steps, and a decay horizon of 18,946 steps. Checkpoints are written every 250 steps. Throughput is about 36 seconds per iteration. A run exceeds the 24-hour scheduling limit and is completed as two serialized allocations, the second resuming from the newest complete checkpoint.

**Replaced slice.** The 72,000 source examples are generated at seed 20260830 with depths 1 to 25 round-robin, branching factor 4, distractor ratio 0.5 and the same generator variant as the controlled study. Seeds are disjoint from those of every evaluation corpus. A per-condition document weight is solved so that the realised synthetic share of loss tokens is exactly ten percent after padding, which is necessary because the three renderings pack with different efficiencies.

**Packing and audit.** Documents are packed by a document-preserving packer at sequence length 8,192 under a fixed shuffle seed. The packer never splits a document; a document exceeding one window is excluded and counted. Tail padding uses a dedicated token masked out of the loss. Overlength and split counts are zero for every condition. Before training begins, a decoded-batch audit reconstructs real loader windows and verifies that no split document is present, that the padding token is masked, and that the realised replacement share matches the target. Training refuses to start if any gate fails.

**Instruction tuning.** Each midtrained checkpoint is converted to the Hugging Face format and instruction-tuned on 100,000 examples of a public instruction corpus at a pinned revision, for one epoch at learning rate $5 \times 10^{-6}$ with warmup ratio 0.03, maximum sequence length 4,096, per-device batch size 1, and seed 3407 (replicate: seed 3408), full-parameter under fully sharded data parallelism with gradient checkpointing. Only the final checkpoint is written.

**Evaluation.** Greedy evaluation uses a standard harness with a paged-attention backend in bfloat16, a maximum model length of 8,192, the model's chat template applied to every prompt, and per-sample logging. The metric is exact match on the extracted answer. The external family is ProofWriter under the open-world assumption, bucketed by proof depth at 0, 1, 2, 3 and 5 with 500 items per depth. The near-transfer family is a held-out sample from the generator at depths 5, 10, 15, 20 and 25 with 200 items per depth, prompted to write a derivation before the answer, with a 2,048-token generation cap. Sampled evaluation draws sixteen completions per item at temperature 0.8 and top-$p$ 0.95 under a fixed seed, replaying the prompts logged by the greedy run, and stops on every end-of-sequence token declared by the checkpoint. The scorer imports the extraction function of the greedy harness.

The general suite is GSM8K, MATH-500, BBH, MMLU, ARC-Challenge, HellaSwag, WinoGrande, PIQA and LogiQA, and the multi-hop tasks are HotpotQA, 2WikiMultihopQA and MuSiQue under standard prompting.

**Statistics.** Contrasts are computed by a paired bootstrap over evaluation documents with 4,000 resamples, resampling document indices once per replicate and applying the same indices to every condition and depth. Reported intervals are the 2.5th and 97.5th percentiles of the replicate distribution.

## J Per-Depth Transfer Tables

Table 6: General-benchmark scores of the five midtrained and instruction-tuned models at seed 3407, in percent. GSM8K is strict-match exact match; MATH-500 is scored by a post-hoc answer-prefix checker; BBH is exact match over the chain-of-thought few-shot suite; the multi-hop tasks report F1 under standard prompting.

| condition | GSM8K | MATH | BBH | MMLU | ARC | HSwag | WinoG | PIQA | LogiQA | Hotpot | 2Wiki | MuSiQue |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Control | 78.6 | 11.8 | 67.4 | 69.8 | 50.3 | 56.3 | 69.8 | 79.3 | 31.2 | 55.7 | 38.9 | 29.9 |
| LongDoc | 79.2 | 12.8 | 68.3 | 69.6 | 51.6 | 56.6 | 70.3 | 79.2 | 29.3 | 57.6 | 35.3 | 30.2 |
| Formal | 79.2 | 12.2 | 68.0 | 69.1 | 52.7 | 56.1 | 70.0 | 79.0 | 28.1 | 57.1 | 36.4 | 27.3 |
| English | 78.6 | 11.4 | 68.3 | 69.2 | 50.3 | 56.0 | 70.4 | 78.9 | 27.8 | 55.9 | 38.2 | 27.2 |
| Condensed | 79.8 | 11.2 | 68.6 | 69.0 | 52.0 | 56.2 | 70.4 | 78.9 | 26.6 | 57.0 | 34.3 | 26.8 |

Table 7: Near-transfer contrasts pooled over inference depths 5 to 25 (1,000 documents), in percentage points, with 95 percent intervals from a paired bootstrap over documents with 4,000 resamples. Intervals cover evaluation sampling error only; the two-seed comparison in Table 8 gives the training-run term.

| contrast | pass@16 | mean single-sample |
|---|---|---|
| English − Control | $+29.4$ $[25.8, 33.1]$ | $+20.5$ $[19.0, 21.9]$ |
| Formal − Control | $+24.4$ $[20.8, 28.0]$ | $+13.7$ $[12.5, 14.9]$ |
| English − LongDoc | $+25.0$ $[21.4, 28.6]$ | $+19.7$ $[18.2, 21.1]$ |
| Formal − LongDoc | $+20.0$ $[16.3, 23.5]$ | $+12.9$ $[11.6, 14.2]$ |
| LongDoc − Control | $+4.4$ $[0.7, 8.0]$ | $+0.8$ $[0.3, 1.4]$ |
| English − Formal | $+5.0$ $[2.1, 7.9]$ | $+6.8$ $[5.2, 8.3]$ |
| Condensed − Control | $+26.5$ $[23.0, 30.1]$ | $+10.6$ $[9.7, 11.5]$ |
| Condensed − LongDoc | $+22.1$ $[18.7, 25.5]$ | $+9.8$ $[8.9, 10.6]$ |
| Condensed − Formal | $+2.1$ $[-0.9, 5.3]$ | $-3.1$ $[-4.5, -1.8]$ |
| English − Condensed | $+2.9$ $[-0.1, 5.8]$ | $+9.9$ $[8.4, 11.3]$ |

The label-bias decomposition of Table 3 also replicates. Pooled over ProofWriter depths 1 to 5 at seed 3408, Control predicts *true* on 65.3 percent of items against 66.3 at seed 3407, and two-class accuracy is 53.5 against 54.4 for Control, 60.4 against 60.3 for Formal, and 68.2 against 68.2 for English.

## K An Earlier Configuration

Before the study of Section 5, the same three-way design was run at sequence length 4,096 with a five percent replacement share, a compact rendering, and derivations capped at depth 14 so that they fit the window, for the same token budget. On a ten-task general suite the three conditions scored 58.2, 58.4 and 58.5 percent macro-average for control, formal and English, and on tagged multi-hop question answering the differences vanished under fallback answer extraction. That configuration had no length-matched control and no deduction benchmark, and its derivations never reached the depths at which the controlled study locates the formal advantage, so it motivated the present design without constraining it.

Table 8: The full deduction suite under both instruction-tuning seeds, greedy exact match in percent. Each pair of columns is the same midtrained checkpoint instruction-tuned at seed 3407 and at seed 3408. ProofWriter uses 500 items per depth and the near-transfer family 200. Over the fifty condition and task pairs the mean absolute difference between seeds is $0.015$ and the largest is $0.050$.

| | CONTROL | | LONGDOC | | FORMAL | | ENGLISH | | CONDENSED | |
|---|---|---|---|---|---|---|---|---|---|---|
| task | 3407 | 3408 | 3407 | 3408 | 3407 | 3408 | 3407 | 3408 | 3407 | 3408 |
| ProofWriter d0 | 44.2 | 46.4 | 45.8 | 45.4 | 48.0 | 48.0 | 48.8 | 48.6 | 46.0 | 44.8 |
| ProofWriter d1 | 32.8 | 33.2 | 32.2 | 32.2 | 35.8 | 36.0 | 38.8 | 39.6 | 34.6 | 34.6 |
| ProofWriter d2 | 44.0 | 43.4 | 44.4 | 43.8 | 49.4 | 49.2 | 56.2 | 56.0 | 46.0 | 46.0 |
| ProofWriter d3 | 44.4 | 43.4 | 45.4 | 44.8 | 50.0 | 50.0 | 57.0 | 57.0 | 48.0 | 48.6 |
| ProofWriter d5 | 46.0 | 44.6 | 44.0 | 43.8 | 50.2 | 50.0 | 56.4 | 55.8 | 47.4 | 44.6 |
| near transfer d5 | 5.0 | 8.0 | 7.5 | 7.5 | 31.5 | 36.5 | 52.5 | 54.5 | 34.5 | 33.0 |
| near transfer d10 | 3.5 | 4.5 | 11.0 | 15.0 | 17.5 | 22.5 | 34.5 | 35.0 | 28.5 | 31.0 |
| near transfer d15 | 11.0 | 11.5 | 12.0 | 10.5 | 27.5 | 23.5 | 30.5 | 25.5 | 18.5 | 22.5 |
| near transfer d20 | 9.0 | 13.0 | 15.0 | 12.5 | 21.5 | 20.0 | 24.5 | 25.5 | 20.5 | 21.5 |
| near transfer d25 | 6.5 | 9.0 | 12.0 | 11.0 | 19.5 | 21.5 | 29.5 | 29.5 | 24.0 | 27.0 |